

Moller scattering 4

Here is another way to compute expected density. Let

$$\begin{aligned}f_{11} &= 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_4)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_4) \\f_{12} &= -32(p_1 \cdot p_2)^2 + 64m^2(p_1 \cdot p_2) \\f_{22} &= 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_3)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_3)\end{aligned}$$

Then

$$\langle |\mathcal{M}|^2 \rangle = \frac{e^4}{4} \left(\frac{f_{11}}{t^2} - \frac{2f_{12}}{tu} + \frac{f_{22}}{u^2} \right)$$

For $m = 0$ we have

$$\begin{aligned}f_{11} &= 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_4)^2 &= 8s^2 + 8u^2 \\f_{12} &= -32(p_1 \cdot p_2)^2 &= -8s^2 \\f_{22} &= 32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_3)^2 &= 8s^2 + 8t^2\end{aligned}$$

Eigenmath script